

Statistics Explained

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methodology - material deprivation by dimension

EU statistics on income and living conditions (EU-SILC) methodology - material deprivation by dimension



This article is part of a [set of articles](#) describing the methodology applied for the computation of the statistical indicators pertinent to the subject area of [Material deprivation \(ilc_mddd\)](#) within the overall domain of [Income and living conditions](#). For these indicators, the article provides a methodological and practical framework of reference. The indicators relevant to the subject area of material deprivation by dimension are the following:

- Material deprivation rate – Economic strain and durables dimension
- Material deprivation rate for the 'Economic strain' dimension
- Mean number of deprivation items among the deprived – Economic strain and durables dimension
- Material deprivation rate for the 'Durables' dimension
- Material deprivation rate for the 'Economic strain' and 'Durables' dimensions
- Material deprivation rate for the 'Housing' dimension
- Material deprivation rate for the 'Environment' dimension
- Severe material deprivation rate

Moreover, since the indicators are of multidimensional structure and can be analysed simultaneously along several dimensions, the separate [datasets](#) providing these indicators along with the different combinations of dimensions are also presented.

Description

- The **material deprivation rate – Economic strain and durables dimension**, refers to persons who are materially deprived, based on their inability to afford to pay: No items - 1 item - 2 items - Fewer than 3 items - 3 items - 3 items or more - Fewer than 4 items - 4 items - 4 items or more - Fewer than 5 items - 5 items - 5 items or more - 6 items - 7 items - 8 items - 9 items from the list of the [material deprivation items](#).

- The **material deprivation rate for the 'Economic strain' dimension**, refers to the percentage of persons who are materially deprived for the 'Economic strain' dimension, based on their inability to afford to pay: No items – 1 item – 2 items – 3 items – 4 items – 5 items from the list of the material deprivation items for the 'Economic strain' dimension. **Materially Deprived for the 'Economic strain' dimension** are characterised those individuals being at the state of economic strain, defined as the state of enforced inability to afford one or more of the following items: (a) to keep home adequately warm; (b) to afford paying for one-week annual holiday away from home; (c) to afford a meal with meat, chicken, fish (or vegetarian equivalent) every second day; (d) to face unexpected financial expenses; (e) to be confronted with payment arrears (mortgage or rent, utility bills, hire purchase instalments or other loan payments).
- The **mean number of deprivation items among the deprived – Economic strain and durables dimension**, is defined as the average number of deprivation items of materially deprived persons.
- The **material deprivation rate for the 'Durables' dimension**, refers to persons in the total population who are materially deprived for the 'Durables' dimension, based on their inability to afford to pay: No items – 1 item – 2 items – 3 items – 4 items – 5 items from the list of the material deprivation items for the 'Durables' dimension. **Materially Deprived for the 'Durables' dimension** are characterised those individuals being at the state of durable strain, defined as the state of enforced lack of one or more of the following items: (a) telephone; (b) colour TV; (c) computer; (d) washing machine; (e) personal car.
- The **material deprivation rate for the 'Economic strain' and 'Durables' dimensions**, refers to the percentage of persons in the total population who are materially deprived for both the 'Economic strain' and 'Durables' dimension, including in the latter the inability to afford a computer, based on their inability to afford to pay: No items – 1 item – 2 items – 3 items – 4 items – 5 items or more from the list of the material deprivation items. The definition of this indicator also covers the enforced inability to afford a computer, as part of the 'Durables' dimension. This item is not included in the list of the nine [material deprivation items](#).
- The **material deprivation for the 'Housing' dimension**, refers to the percentage of persons in the total population who are materially deprived for the 'Housing' dimension, based on the dwelling problems suffered from: No items – 1 item – 2 items – 3 items – 4 items from the list of the material deprivation items. **Materially Deprived for the 'Housing' dimension** are characterised those individuals being at the state of housing deprivation, defined as the state of suffering one or more of the following dwelling problems: (a) leaking roof / damp walls / floors / foundation or rot in window frames; (b) accommodation too dark; (c) no bath/shower; (d) no indoor flushing toilet for sole use of the household.
- The **material deprivation for the 'Environment' dimension**, refers to the percentage of persons in the total population who are materially deprived for the 'Environment' dimension, based on the dwelling problems suffered from: No items – 1 item – 2 items – 3 items from the list of the material deprivation items. **Materially Deprived for the 'Environment' dimension** are characterised those individuals being at the state of environmental deprivation,

defined as the state of being affected by the dwelling's local environment, with reference to the following three items: (a) noise from neighbours or from the street; (b) pollution, grime or other environmental problems; (c) crime, violence or vandalism in the area.

- [Severe material deprivation rate](#)

Statistical population

For the material deprivation rate for all dimensions, i.e. economic strain and durables dimension, economic strain dimension, durables, economic strain and durables dimensions, housing dimension and environment dimension, the statistical population consists of all persons living in private households, excluding those persons with missing values for any of the nine material deprivation items in each dimension or for any of the [dimensions](#) with which the indicator is disseminated.

With regard to the calculation of the mean number of deprivation items among the deprived (economic strain and durables dimension), all persons living in private households that are considered [materially deprived](#) are included in the reference population. People with missing values for any of the nine deprivation items, or for any of the [dimensions](#) with which the indicator is disseminated are excluded from the calculation.

The severe material deprivation rate refers to all persons living in private households. People with missing values for any of the nine material deprivation items or any of the [dimensions](#) with which the indicator is disseminated are excluded from the calculation. However, the indicator covers subsets of population when presented along with different dimensions. More specifically, it covers the population aged 18 and over when broken down by most frequent activity status, level of education, broad group of citizenship, broad group of country of birth. Additionally, when calculated for children (i.e. severe material deprivation rate for children by highest education level of their parents), it refers to the population aged 0 to 17 living in private households.

In any case, persons living in [collective households and in institutions](#) are generally excluded from calculations.

Reference period

All indicators are collected and disseminated on an annual basis and refer to the survey year.

The reference period for all dimensions along with the indicators are disseminated as well as variables related to the items in question is the survey year, except for income, age and variables on arrears. For income and activity status, the reference period is the income reference period. The income reference period is a fixed 12-month period (such as the previous calendar or tax year) for all countries except the United Kingdom, for which the income reference period is the current year, and Ireland, for which the survey is continuous and income is collected for the last twelve months. As far as age is concerned, it refers to the age of the respondent at the end of the income reference period. Regarding variables on arrears, these refer to the last 12 months.

Unit of Measurement

The number of persons who are materially deprived for the economic strain and durables dimension and the number of persons severely materially deprived (in thousands) is provided.

The material deprivation for all other dimensions (i.e. economic strain dimension, durables, economic strain and durables dimensions, housing dimension and environment dimension) is made available as a percentage.

The mean number of deprivation items among the deprived – Economic strain and durables dimension is given as a (average) number of items.

Dimensions

The separate datasets provide each indicator along with the [Geopolitical entity](#) and [time](#) dimensions and the dimensions presented below.

The material deprivation rate - economic strain and durables dimension is presented along with the following dimensions:

- age group, sex and number of deprivation items

The mean number of deprivation items among the deprived - Economic strain and durables dimensions is presented along with the following dimensions:

- age group and sex

The material deprivation rate for the 'Durables' dimension is presented along with the following dimensions:

- household type and income group (INCGRP) (total, below 60 % of median equivalised income, above 60 % of median equivalised income), number of deprivation items

The material deprivation rate for the 'Economic strain' and 'Durables' dimensions are presented broken down by:

- household type and income group (INCGRP) (total, below 60 % of median equivalised income, above 60 % of median equivalised income), number of deprivation items

The material deprivation rate for the 'Housing' dimension is presented broken down by:

- household type and income group (INCGRP) (total, below 60 % of median equivalised income, above 60 % of median equivalised income), number of deprivation items

The material deprivation rate for the 'Environment' dimension is presented broken down by:

- household type and income group (INCGRP) (total, below 60 % of median equivalised income, above 60 % of median equivalised income), number of deprivation items

The severe material deprivation rate is disseminated along with the following dimensions:

- age group and sex
- age group, sex and most frequent activity status (population aged 18 and over)
- income quintile and household type

- age group, sex and educational level [\(ISCED\)](#) (population aged 18 and over)
- age group, sex and broad group of citizenship (population aged 18 and over)
- age group, sex and broad group of country of birth (population aged 18 and over)
- tenure status
- NUTS 2 regions
- degree of urbanisation [\(DEGURBA\)](#)

Especially when the indicator is computed for children, i.e. severe material deprivation rate for children (population aged 0 to 17 years), it is broken down in the following dimensions:

- highest education level of parents (ISCED) (population aged 0 to 17 years)

Calculation method

1. Material deprivation rate – Economic strain and durables dimension:

Let N_ITEM be the total number of items that a person cannot afford to pay, with N_ITEM ranging from 0 to 9. Material deprivation rate broken down by each combination of [dimensions](#) (k) ($DEPR_TOTL_{at_k}$) is calculated as the percentage of people (or thousands of people) in each dimension who cannot afford to pay N_ITEM of 'material deprivation items'.

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$DEPR_TOTL_{at_k} = \frac{\sum_{\forall i_{at_k} where \Lambda} \sum_{\forall i_{at_j}} \sum_{\forall i_{at_k} where \Lambda}}{\sum_{\forall i_{at_k} where \Lambda}}$$

where j takes the values: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, Less than 3 (i.e. 0-2), Less than 4 (i.e. 0-3), Less than 5 (i.e. 0-4), More than 3 (i.e. 3-9), More than 4 (i.e. 4-9) and More than 5 (i.e. 5-9).

2. Mean number of deprivation items among the deprived – Economic strain and durables dimension:

Let N_ITEM be the total number of items that a person cannot afford to pay. Mean number of deprivation items broken down by each combination of dimensions (k) ($MEAN_TOTL_{at_k where N_ITEM \geq 3}$) ($N_ITEM \geq 3$) is calculated as the weighted average number of items lacked by persons in each k who cannot afford to pay at least 3 of 'material deprivation items'.

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$MEAN_TOTL_{at_k where N_ITEM \geq 3} = \frac{\sum_{\forall i_{at_k} where N_ITEM \geq 3} \sum_{\forall i_{at_k} where N_ITEM \geq 3}}{\sum_{\forall i_{at_k} where N_ITEM \geq 3}}$$

3. Material deprivation for the 'Economic strain' dimension:

Let N_ITEM be the total number of items in the 'Economic strain' dimension that a person cannot afford to pay, with N_ITEM ranging from 0 to 5. Material deprivation for the 'Economic strain' dimension broken down by each combination

of dimensions ($DEPR_ECON_{at_k}$) is calculated as the percentage of people in each dimension who cannot afford to pay N_ITEM of [material deprivation items for the 'Economic strain' dimension](#).

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$DEPR_ECON_{at_k} = \frac{\sum_{\forall i_{at_k} \text{ where } j} \sum_{\forall i_{at_k}} 1}{\sum_{\forall i_{at_k}} 1}$$

where j = 0, 1, 2, 3, 4, 5 denotes the number of deprivation items for the 'Economic strain' dimension.

4. Material deprivation for the 'Durables' dimension:

Let N_ITEM be the total number of items in the 'Durables' dimension that a person cannot afford to pay, with N_ITEM ranging from 0 to 5. Material deprivation for the 'Durables' dimension broken down by each combination of dimensions (k) ($DEPR_DUR_{at_k}$) is calculated as the percentage of people in each k who cannot afford to pay N_ITEM of [material deprivation items for the 'Durables' dimension](#).

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$DEPR_DUR_{at_k} = \frac{\sum_{\forall i_{at_k} \text{ where } N=1} \sum_{\forall i_{at_k}} 1}{\sum_{\forall i_{at_k}} 1}$$

where j = 0, 1, 2, 3, 4, 5 denotes the number of deprivation items for the 'Durables' dimension.

5. Material deprivation for the 'Economic strain' and 'Durables' dimensions:

Let N_ITEM be the total number of items that a person cannot afford to pay, with N_ITEM ranging from 0 to 10. Material deprivation for the 'Economic strain' and 'Durables' dimension broken down by each combination of dimensions (k) ($DEPR_TOTL_{at_k}$) is calculated as the percentage of people in each (k) who cannot afford to pay N_ITEM of [material deprivation items](#).

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$DEPR_TOTL_{at_k} = \frac{\sum_{\forall i_{at_k} \text{ where } N} \sum_{\forall i_{at_k}} 1}{\sum_{\forall i_{at_k}} 1}$$

where j = 0, 1, 2, 3, 4, 5 items or more denotes the number of material deprivation items as defined [above](#).

6. Material deprivation for the 'Housing' dimension:

Let N_ITEM be the total number of housing items (dwelling problems) in the 'Housing' dimension that a person is deprived of, with N_ITEM ranging from 0 to 4. Material deprivation for the 'Housing' dimension broken down by each combination of dimensions (k) ($DEPR_HOUS_{at_k}$) is calculated as the percentage of people in each k who are deprived of N_ITEM from the list of [housing deprivation items](#).

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$DEPR_HOUS_{at_k} = \frac{\sum_{\forall i_{at_k} \text{ where } l} \sum_{\forall i_{at_k}} 1}{\sum_{\forall i_{at_k}} 1}$$

where $j = 0, 1, 2, 3, 4$ denotes the number of deprivation items for the 'Housing' dimension as defined above.

7. Material deprivation for the 'Environment' dimension:

Let N_ITEM be the total number of items (dwelling problems) in the 'Environment' dimension from which a person suffers, with N_ITEM ranging from 0 to 3. Material deprivation for the 'Environment' dimension broken down by each combination of dimensions (k) ($DEPR_ENV_{at_k}$) is calculated as the percentage of people in each k who suffer from N_ITEM of [environmental deprivation items](#).

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$DEPR_ENV_{at_k} = \frac{\sum_{\forall i_{at_k} \text{ where } N_I} \frac{1}{N_I}}{\sum_{\forall i_{at_k}} 1}$$

where $j = 0, 1, 2, 3$ denotes the number of deprivation items for the 'Environment' dimension as defined [above](#).

8. Severe material deprivation rate:

Let N_ITEM be the total number of items that a person of the population, or a [subset of population](#), cannot afford to pay, with N_ITEM ranging from 4 to 9. Severe material deprivation rate broken down by each combination of dimensions (k) ($SEV_DEPR_TOTL_{at_k}$) is calculated as the percentage of people (or thousands of people) in each k who cannot afford to pay at least four of 'material deprivation items' ($N_ITEM \geq 4$).

The weight variable used is the Adjusted Cross Sectional Weight (RB050a).

$$SEV_DEPR_TOTL_{at_k} = \frac{\sum_{\forall i_{at}} 1}{N_{at}}$$

$$SEV_DEPR_TOTL_{at_k} = \frac{\sum_{\forall i_{at}} 1}{N_{at}}$$

where j takes the values: 4, 5, 6, 7, 8, 9.

With regard to the calculation of the above indicators, the following methodological issues should be taken into consideration:

- The threshold of four items to depict severe material deprivation has been chosen for a mixture of empirical and practical reasons since a previous threshold of 3 items had resulted in excessively high, and politically unmanageable, estimates of levels of deprivation across the EU.
- Accuracy should be considered as restricted despite the overall specifications contained in the legislative framework (Regulation (EC) No 1177/2003):
- Existence of a compulsory minimum sample size for each country;
- Imputation of missing and inaccurate values.
- Comparability of material deprivation indicators between countries is also an issue for consideration. A process of harmonization of material deprivation variables is in progress and aims at improving the comparability of the material deprivation statistics across countries by harmonizing the wording of the EU guidelines, the national questionnaires as well as the national interview guidelines.
- The transition towards more harmonized statistics along – for some countries - with the implementation of new questions

created some breaks over time. Most changes occur between the 2007 and 2008 operation.

- It is foreseen that the list of items supporting this indicator will be reviewed by 2015.

Moreover, there are some methodological limitations that pertain to the following dimensions accompanying the indicators: Age, Activity status, Citizenship, Country of birth, Degree of urbanisation, Educational level, Highest educational level of parents, Household type, NUTS region, Tenure status.

Main concepts used

For the production of the indicators relevant to the subject area of material deprivation by dimension, the variables listed below are also involved in computations:

Income quantile, Material deprivation (MD), Median equivalised disposable income (MEDIAN 20).

SAS program files

SAS programming routines developed for the computation of the EU-SILC material deprivation by dimension datasets along with the different dimensions are listed below.

Dataset	SAS program file
Material deprivation rate for the 'Economic strain' and 'Durables' dimensions, by number of items of deprivation (ilc_sjp8)	OV9b.sas
Mean number of deprivation items among the deprived - Economic strain and durables dimension (ilc_sis4)	OV9b.sas
Material deprivation for the 'Economic strain' dimension, by number of items of deprivation (ilc_mddd01)	_mddd01.sas
Material deprivation for the 'Durables' dimension, by number of items of deprivation (ilc_mddd02)	_mddd02.sas
Material deprivation for the 'Economic strain' and 'Durables' dimensions, by number of items of deprivation (ilc_mddd03)	_mddd03.sas
Material deprivation for the 'Housing' dimension, by household type, poverty status and by number of items of deprivation (ilc_mddd04a)	_mddd04.sas
Material deprivation for the 'Housing' dimension, by age, sex and by number of items of deprivation (ilc_mddd04b)	_mddd04.sas
Material deprivation for the 'Environment' dimension, by number of items of deprivation (ilc_mddd05)	_mddd05.sas
Severe material deprivation rate by age and sex (ilc_mddd11)	mddd11.sas
Severe material deprivation rate by most frequent activity status (population aged 18 and over) (ilc_mddd12)	mddd12.sas
Severe material deprivation rate by income quintile and household type (ilc_mddd13)	mddd13.sas
Severe material deprivation rate by educational attainment level (population aged	mddd14.sas

18 and over) (ilc_mddd14)	
Severe material deprivation rate by broad group of citizenship (population aged 18 and over) (ilc_mddd15)	mddd15.sas
Severe material deprivation rate by broad group of country of birth (population aged 18 and over) (ilc_mddd16)	mddd16.sas
Severe material deprivation rate by tenure status (ilc_mddd17)	mddd17.sas
Severe material deprivation rate by NUTS 2 regions (ilc_mddd21)	mddd21.sas
Severe material deprivation rate by degree of urbanisation (ilc_mddd23)	_mddd23.sas
Severe material deprivation rate for children by educational attainment level of their parents (population aged 0 to 17 years) (ilc_mddd60)	mddd60.sas

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